Responsibility Drift in AI-Mediated Systems

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I. Research Background

With the rapid development of artificial intelligence technologies, an increasing number of organizations have begun to rely on AI systems for complex decision support. AI systems are capable not only of processing large volumes of data, but also of generating predictive outputs through machine learning models and, in some cases, directly participating in decision execution. In areas such as corporate

governance, financial transactions, medical diagnosis, autonomous driving, and public infrastructure management, AI systems are gradually becoming part of critical decision structures. As system complexity continues to increase, decision processes have shifted from traditional single-subject models toward multi-node collaborative structures.

In such environments, decision chains are typically composed of multiple technical and organizational layers. For example, data teams are responsible for collecting and cleaning data, algorithm teams are responsible for model design and training, system engineering teams are responsible for deployment and operation, and organizational management applies system outputs to concrete business decisions. At the same time, automated execution systems may directly carry out system outputs without human intervention. This complex decision structure, jointly constituted by technical nodes and organizational nodes, is referred to in this paper as an **AI-mediated system**.

In traditional decision structures, responsibility is usually concentrated in clearly identifiable human actors. Decision-makers formulate decisions, executors implement them, and organizational structures ensure that responsibility can be clearly identified along the decision chain. Within such structures, responsibility occupies a relatively stable position and usually corresponds to the authority structure. When problems arise, responsibility can be traced through organizational hierarchy.

However, in AI-mediated systems, this responsibility structure is undergoing significant change. As the participation of automated systems increases, system execution capacity is greatly enhanced, while responsibility-bearing structures gradually become blurred. Different nodes each assume partial functions within system operation, while overall system behavior is jointly produced by multiple nodes. When failures occur, each node may explain only its partial role, while the system as a whole lacks a clearly identifiable bearer of responsibility.

This structural transformation causes responsibility to shift gradually from a concentrated mode to a distributed mode. Responsibility no longer remains stably concentrated in a single node, but may gradually diffuse within the system structure. The stronger the execution capacity, the greater the number of nodes, and the higher the degree of automation, the more pronounced this transformation becomes.

In some cases, systems may continue to operate with high efficiency, while responsibility structures gradually lose their stable positions. Responsibility does not disappear; rather, it is gradually displaced within complex system structures,

making it difficult to form a clear relation of responsibility-bearing within the system.

This paper refers to this phenomenon as **Responsibility Drift**. Responsibility drift does not mean that there is no responsibility in the system, nor does it mean that no human actors participate in decision processes. Rather, it describes a structural transformation: when decisions are jointly produced by multiple technical and organizational nodes, responsibility may gradually diffuse within the system structure and lose its stable concentrated position.

Against the backdrop of the continued expansion of AI systems, this transformation of responsibility structures is becoming an important issue in complex systems governance. As AI systems exert growing influence over social and economic structures, understanding responsibility drift is of significant importance for AI governance, complex systems stability, and decision-structure design.

Accordingly, this study approaches the problem from the perspective of complex system structures, analyzes the formation mechanisms of responsibility drift in AI-mediated systems, and seeks to explain how responsibility structures gradually shift from concentrated to distributed states in multi-node decision environments.

II. Research Questions

In complex system environments, responsibility structures are not merely matters of organizational management; they are also matters of system structure. As artificial intelligence systems increasingly participate in decision processes and execute decision behavior through multi-node technical architectures, traditional responsibility structures centered on single actors may become difficult to sustain. Under such conditions, responsibility no longer remains stably concentrated in one node, but may gradually diffuse across multiple system nodes.

With the spread of AI-mediated systems, more and more critical decisions are jointly produced by technical and organizational nodes. Data systems provide foundational information for decisions, algorithmic models perform prediction or computation, technical systems execute operational processes, and organizational layers apply outputs in actual decision environments. In such structures, system behavior is often jointly generated by multiple nodes rather than by a single subject.

This multi-node structure substantially alters the position of responsibility within the system. In traditional organizational structures, responsibility typically corresponds to authority, and relatively stable relations exist between decision

subjects and responsibility-bearing subjects. In AI-mediated systems, however, different nodes may each assume partial roles while overall system outcomes are jointly formed. As a result, the position of responsibility within the system structure becomes increasingly unstable.

The core question addressed in this paper is therefore: how does responsibility gradually drift within multi-node decision structures in AI-mediated systems, and how does this structural transformation affect the stability of complex systems?

More specifically, this study seeks to answer three interrelated questions. First, how does responsibility move from a concentrated state toward a distributed state in multi-node decision systems? When systems are jointly constituted by technical and organizational nodes, the concentrated position of responsibility may be gradually weakened, resulting in fragmented responsibility structures.

Second, why are responsibility chains prone to structural discontinuity in highly automated environments? When system outputs enter execution directly through automated processes, human actors may fail to assume responsibility-bearing roles at key decision points. Under such circumstances, discontinuities may emerge within the responsibility chain, making it difficult for responsibility to retain a stable position in the decision structure.

Third, what effects may responsibility drift have on complex systems governance? As responsibility gradually diffuses within system structures, organizational governance structures, risk-control mechanisms, and system stability may all be affected. Understanding this structural transformation is therefore important for AI governance and complex systems management.

Through the analysis of these questions, this paper seeks to construct a structural framework for explaining how responsibility drift emerges in AI-mediated systems and how this structural transformation affects responsibility-bearing mechanisms in complex systems. This analysis not only contributes to the understanding of responsibility structures in AI systems, but also offers a new theoretical perspective for future research on complex systems governance.

On this basis, the paper further connects the concept of responsibility drift with the author's earlier structural theory of judgment in complex systems. In *Judgment Before Momentum*, human judgment was treated as an important condition for maintaining structural stability. In *Human Judgment as an Irreducible Interface in High-Complexity Systems*, human judgment was defined as an irreducible interface within complex systems. In *Human Responsibility Node in High-Replication Systems*,

the responsibility node was introduced to describe the concentrated position of responsibility within system structures.

Building on these earlier studies, this paper further argues that when responsibility nodes are absent or structurally weakened, responsibility may gradually drift within the system structure, thereby giving rise to the phenomenon discussed here as **Responsibility Drift**.

III. Structural Features of AI-Mediated Systems

AI-mediated systems are typically constituted by multiple technical and organizational layers. Such systems are not single technical components, but complex structures jointly formed by data infrastructures, algorithmic models, technical execution systems, and organizational decision arrangements. As artificial intelligence technologies have become widely applied in enterprise operations, platform governance, and social infrastructure, this multi-layered structure has increasingly become a defining feature of modern decision systems.

In a typical AI-mediated system, the structure can be divided into four basic layers. The first is the **Data Layer**, which is responsible for data collection, storage, and preprocessing. The data layer provides the foundational information for system operation, and its quality and completeness directly affect the accuracy of system decisions. The second is the **Model Layer**, which is responsible for algorithmic computation and prediction. Machine learning models analyze and train on data to generate predictive outputs, which then serve as inputs into decision processes. The third is the **System Layer**, which is responsible for technical integration and automated execution. This layer typically transforms model outputs into executable operations through software architecture, API interfaces, and automated processes. The fourth is the **Organizational Layer**, which applies system outputs to concrete business decisions or governance actions.

Within this structure, outputs are often generated by algorithmic models and then enter the execution stage through automated processes. Based on model predictions, systems may automatically complete tasks such as recommending content, executing trades, adjusting parameters, or triggering specific operations. In some cases, human actors participate only during system design, without direct participation in execution once the system is running.

Compared with traditional decision structures, this architecture significantly changes the position of human actors in the decision chain. In traditional organizational settings, decisions are made by clearly identifiable actors, and responsibility and authority remain relatively stable. In AI-mediated systems, however, decision processes are decomposed into multiple technical steps, each performed by different nodes. Data teams handle data processing, algorithm teams develop models, system engineers deploy the systems, and organizational layers apply outputs.

Decision-making therefore no longer appears as the action of a single subject, but rather as the joint participation of multiple nodes. System behavior emerges from interactions among technical and organizational nodes. In such environments, system efficiency may increase substantially, while the structural position of responsibility becomes more complex.

As system scale expands, this complexity becomes even more pronounced. In large AI systems, models may be replicated across multiple application scenarios, system execution may span multiple technical platforms, and organizational decision structures may involve multiple departments or institutions. System behavior thus increasingly takes on a distributed rather than centralized form.

This distributed structure causes the position of responsibility within the system to change gradually. Different nodes assume partial functions in system operation, while overall system results are jointly produced by multiple nodes. When problems arise, responsibility becomes difficult to locate in a clearly identifiable structural position.

Understanding the structural features of AI-mediated systems is therefore crucial for analyzing responsibility drift. Responsibility drift is not merely a matter of organizational management; it is a consequence of structural transformation in complex systems.

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┌──────────────────────────────────────────────┐
│          Organizational Layer                │
│      Human Decision and Governance           │
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┌──────────────────────────────────────────────┐
│               System Layer                   │
│      Automation / Software Execution         │
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┌──────────────────────────────────────────┐
│                Model Layer               │
│         AI Models / Algorithmic Computation  │
└──────────────────────────────────────────┘

                      ▲
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┌──────────────────────────────────────────┐
│                 Data Layer               │
│          Data Collection / Data Input    │
└──────────────────────────────────────────┘


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Figure 1. Architecture of AI-Mediated Decision Systems

AI-mediated decision systems typically consist of multiple structural layers, including the data layer, model layer, system layer, and organizational decision layer. Together, these layers form distributed decision environments in which technical components and organizational actors interact.

Source: Author's conceptual framework.

IV. Responsibility Structures in Complex Systems

Complex systems theory generally understands systems as dynamic structures composed of multiple interacting nodes. In such systems, overall behavior is not directly controlled by a single subject, but is instead jointly produced by interactions among multiple nodes. Information exchange, feedback mechanisms, and structural couplings among nodes often determine the evolutionary pathways of system behavior.

System behavior in complex environments typically has emergent features. That is, overall system behavior cannot be simply explained by the behavior of a single node, but instead emerges from interactions among multiple nodes. As system scale increases and node numbers grow, structures tend to exhibit higher complexity.

In traditional organizational structures, responsibility generally corresponds to authority. Decision-makers formulate decisions, executors implement them, and organizational hierarchies ensure that responsibility can be clearly identified along decision chains. When outcomes arise, responsibility can usually be traced through organizational structure.

Such responsibility structures have two important features. First, responsibility is usually concentrated in clearly identifiable subjects. Decision-makers possess authority and assume corresponding responsibility. Second, the responsibility chain is relatively stable. Clear relations between decision and execution allow responsibility to retain continuity within organizational structures.

However, as decision processes move into complex system environments, traditional responsibility structures may change. As the number of nodes increases, decision behavior shifts from single-subject action to multi-node coordination. In this setting, system behavior is no longer directly generated by one actor, but by interactions among multiple nodes.

Different nodes in complex systems often perform different functions. Some process information, some support decision-making, and others execute operations. Overall system behavior emerges from the coordinated operation of these nodes rather than from any single subject.

As interactions among nodes increase, the position of responsibility within the system may also change. When system behavior is jointly produced by multiple nodes, responsibility no longer remains stably concentrated in a single subject, but may instead take on distributed forms across multiple nodes.

This structural transformation is especially visible in technical systems. As AI systems and automation increasingly participate in decision processes, system structures become further complicated. More tightly coupled relations among technical components and organizational structures make the position of responsibility increasingly difficult to identify.

Thus, in complex systems, responsibility structure is no longer merely a matter of organizational management; it is also a system-structural issue. The position of responsibility is closely tied to system structure and may shift as structures evolve.

Understanding responsibility structures in complex systems is therefore essential for analyzing responsibility drift in AI-mediated environments. As system structures gradually shift from concentrated to distributed forms, the position of responsibility may likewise shift.

V. Historical Patterns of Responsibility Structures

Throughout social development, responsibility structures have always been closely linked to technological and organizational structures. As modes of production and information processing change, decision structures likewise evolve. Different technological stages tend to generate different forms of responsibility.

In the industrial era, responsibility structures were typically organized around stable hierarchies. Industrial production systems centered on factories and firms, with clearly layered authority structures. Decision power was usually concentrated at the management level, while subordinate employees implemented tasks. In such structures, decision and execution were clearly related. Responsibility corresponded to organizational authority, and when problems occurred, responsibility could be traced through hierarchical structure. Because decision subjects were relatively concentrated, the position of responsibility was usually quite clear.

With the development of information technology, organizational structures gradually changed. In the information age, improved information-processing capacity allowed organizational coordination to shift from hierarchical forms toward network forms. Information systems connected departments, organizations, and regions, so that decision processes no longer depended exclusively on single hierarchies.

In such environments, decision structures began to exhibit networked characteristics. Multiple nodes could simultaneously participate in information processing and decision formation. For example, in enterprise information systems, data systems, business systems, and management systems formed complex networks of information exchange. Different nodes played different roles in decision processes, shifting decision behavior gradually from single-subject action toward multi-node collaboration.

As decision structures became increasingly networked, responsibility structures also became more complex. Responsibility was no longer borne exclusively by single actors, but could be distributed across multiple organizational nodes. Although

responsibility could still be traced through organizational structures, the responsibility chain became increasingly multi-layered.

With the arrival of the AI era, this structural complexity increased further. Artificial intelligence systems can not only process large amounts of information, but also generate predictive outputs and participate directly in decision processes. In many applications, outputs may even enter execution through automated processes. AI systems therefore become critical nodes in decision chains.

In AI-mediated systems, decision structures are no longer composed exclusively of human actors, but jointly by technical and organizational nodes. Data systems, algorithmic models, software architectures, and organizational management layers become highly coupled. System behavior is often jointly produced by multiple nodes rather than by single subjects.

As these multi-node structures expand, the position of responsibility may also change gradually. Different nodes assume local responsibilities for system operation, while overall system behavior is jointly formed. As system scale grows, the concentrated position of responsibility may gradually be weakened.

Thus, from the industrial era through the information age and into the AI era, responsibility structures have evolved gradually from concentrated forms toward distributed forms. As complexity increases, responsibility structures shift from stable hierarchical relations to more complex network forms.

Within this historical evolution, the position of responsibility becomes increasingly difficult to identify. As technical nodes increasingly participate in or even dominate decision processes, responsibility structures may gradually be displaced within the system. This structural transformation provides the background for the phenomenon of responsibility drift discussed in this paper.

VI. Systemic Transformation of Responsibility Structures

As artificial intelligence increasingly enters organizational decision systems, decision structures undergo important transformation. In traditional organizational settings, decisions were usually made by human actors, while technical systems primarily served as tools for information processing. In AI-mediated systems, however, technical systems no longer function merely as auxiliary instruments; they become important components of decision chains.

Within such systems, decision processes are jointly completed by multiple nodes. Data systems collect and process information, algorithmic models generate predictions or recommendations, software systems integrate technical components and execute automated procedures, and organizational layers apply system outputs to real decision environments. Different nodes assume different roles and together form complex decision structures.

As technical systems increasingly participate in decision processes, decision chains become distinctly distributed. System behavior is no longer entirely controlled by one subject, but jointly formed by interactions among multiple nodes. In such structures, each node often assumes only partial functions, while overall decisions emerge from the combination of these partial behaviors.

This transformation has important implications for responsibility structures. In traditional organizations, responsibility was concentrated in actors who possessed decision authority. When results emerged, responsibility could be identified through hierarchy. In distributed decision systems, however, decision processes are completed jointly by multiple nodes, and responsibility no longer remains stably concentrated in a single actor.

When system behavior is jointly produced by multiple nodes, each node often performs only partial functions. A data node provides data sources, an algorithmic node generates computational outputs, a system node executes automated processes, and an organizational node applies outputs. Under such conditions, system behavior is formed in a distinctly distributed way.

As distributed structures expand, the degree of concentration of responsibility may gradually decline. Different nodes perform local tasks, but overall behavior is produced by the aggregation of these tasks. As system scale increases, the position of responsibility may become increasingly difficult to identify.

Automation further intensifies this transformation in AI-mediated systems. Outputs may quickly enter execution through automated processes, while human actors may participate only at the level of system design or oversight. Under such conditions, responsibility structures in decision chains may change even more.

When execution behavior is largely performed by automated systems and different nodes only assume local functions, responsibility may gradually shift from concentrated states toward distributed states. This transformation does not mean that responsibility disappears, but rather that its position within the system changes.

Thus, in AI-mediated systems, responsibility structures gradually move from concentrated to distributed modes. Different nodes assume different duties, while overall system behavior is jointly produced. As system scale and complexity increase, the concentrated position of responsibility may gradually weaken, producing new structural states.

This systemic transformation provides important conditions for responsibility drift. When responsibility is no longer stably concentrated in one position, but distributed across multiple nodes, its position within decision chains may gradually diffuse or shift. This paper refers to this structural phenomenon as responsibility drift.

VII. The Concept of Responsibility Drift

This paper introduces the concept of **Responsibility Drift** to describe the structural phenomenon in which responsibility gradually diffuses and shifts within AI-mediated systems. In complex decision systems, when decision behavior is jointly completed by multiple nodes, responsibility may no longer remain stably concentrated in a single subject, but may move gradually among different nodes as system structures change.

Responsibility drift does not mean that there is no responsibility in the system. Rather, it means that responsibility gradually loses a clearly identifiable responsibility-bearing position within the system structure. In such cases, system behavior is still jointly produced by multiple nodes, but the concentrated location of responsibility within the decision chain becomes difficult to identify.

In traditional decision structures, responsibility is usually concentrated in actors who hold decision authority. Decision-makers formulate decisions, executors carry them out, and responsibility chains can be traced through organizational structure. Because there is a stable correspondence between decision behavior and authority structure, responsibility generally remains clear.

In AI-mediated systems, however, decision processes are typically completed jointly by multiple technical and organizational nodes. Data systems provide informational inputs, algorithmic models generate computational outputs, software systems execute automated processes, and organizational structures apply outputs in practical environments. Different nodes assume different functions within the structure.

When system behavior is jointly produced by multiple nodes, responsibility no longer remains stably concentrated in one node. Each node assumes only local tasks,

while overall results emerge from the combination of these partial functions. Under these conditions, responsibility may gradually move from concentrated states toward distributed states.

As system complexity increases, the position of responsibility may change further. Different nodes assume responsibilities for data processing, model computation, system execution, and organizational decision-making. Because these functions are distributed across different layers and subjects, the concentrated position of responsibility may gradually weaken.

Responsibility therefore no longer appears as the stable burden of a single subject, but may gradually diffuse during system operation. Different nodes each assume local tasks, yet overall system behavior is jointly formed. When consequences arise, the structural location of responsibility may become difficult to identify.

Responsibility drift is therefore not the disappearance of responsibility, but a process of structural transformation. Responsibility moves from relatively concentrated states toward more distributed states. As the number of nodes grows and the degree of automation increases, this transformation may repeatedly appear in complex systems.

Responsibility drift can be understood as a structural state of systems. When responsibility-related variables are distributed across different nodes rather than intersecting in one stable position, the degree of concentration of responsibility declines. System behavior can continue, but the identifiability of responsibility structures diminishes.

In AI-mediated systems, the emergence of responsibility drift is closely tied to system structure. As decision processes increasingly take the form of multi-node coordination, the concentrated position of responsibility may gradually weaken. When this structural transformation continues, responsibility drift becomes an important phenomenon in complex decision systems.

The concept of responsibility drift therefore provides a structural perspective for understanding changes in responsibility structures in AI-mediated systems. By treating responsibility as a structural variable rather than merely a normative concept, it becomes possible to explain how responsibility structures shift as technical and organizational structures evolve.

VIII. Conditions for the Emergence of Responsibility Drift

Responsibility drift is not typically caused by a single factor, but emerges gradually under specific system-structural conditions. In complex decision systems, when structures simultaneously exhibit multi-node decision-making, separation of technical layers, and automated execution capacities, the concentrated position of responsibility may gradually change. Under such structural conditions, responsibility structures are more likely to shift from concentrated modes toward distributed modes, thereby giving rise to responsibility drift.

First, multi-node decision structures are one important condition. In traditional decision environments, decisions are usually made by a single actor, allowing responsibility to be concentrated in a clearly identifiable location. In complex systems, however, decisions are often jointly completed by multiple nodes. Different nodes perform different functions, such as data processing, algorithmic computation, system execution, and organizational decision-making. When system behavior is jointly generated by multiple nodes, each node assumes only partial duties, making responsibility less likely to remain stably concentrated.

Second, the separation of technical layers is another important condition. In AI-mediated systems, structures usually consist of multiple technical layers such as data, models, systems, and applications. Different layers are responsible for data collection, algorithmic computation, system integration, and business application. Because these functions are handled by different teams or technical components, the process through which system behavior is formed becomes stratified. When consequences arise, it becomes difficult to maintain clear correspondences between responsibility and different layers, which promotes structural diffusion.

Third, automated execution further intensifies the likelihood of responsibility drift. In traditional decision structures, execution behavior is usually performed by human actors, allowing responsibility to remain directly connected to execution. In automated environments, however, outputs can directly enter execution through software processes or algorithmic mechanisms. Automation reduces direct human participation in execution and enables decision results to propagate quickly within technical structures. Under such conditions, the relation between execution behavior and responsible actors becomes increasingly blurred.

In addition, system replication capacity is another important factor. In contemporary technical environments, algorithmic models, software systems, and data structures often possess strong replication capacity. Outputs can propagate rapidly across different nodes and be repeatedly used in multiple environments. Responsibility structures, however, do not automatically expand with system

replication. As outputs continue to diffuse across multiple environments, the concentrated position of responsibility may be gradually weakened.

When these structural conditions occur together, the position of responsibility may progressively shift. Multi-node decision structures cause system behavior to be jointly produced by multiple nodes, the separation of technical layers makes the production of behavior more complex, and automation and system replication accelerate the pace of execution diffusion. Through the interaction of these factors, the concentrated position of responsibility may gradually weaken, giving rise to responsibility drift.

Responsibility drift should therefore be understood as a structural system state rather than a singular behavioral result. It emerges under specific structural conditions in complex systems and becomes more visible as system scale and technical capability increase. Understanding these conditions helps explain how responsibility structures change in AI-mediated systems under multi-node environments.

IX. Mechanisms of Responsibility Fragmentation

In multi-node systems, different actors typically assume different roles. Data teams are responsible for data collection and quality management, algorithm teams for model design and training, system teams for technical deployment and platform operation, and organizational management for applying outputs to concrete business decisions. Under such conditions, each node assumes only local functions, while overall system outcomes are jointly produced by multiple nodes.

This multi-node structure gives decision processes a distinctly differentiated character. Different nodes are responsible for different parts of system operation, such as information processing, model computation, system execution, and business application. Because these tasks differ technically and organizationally, nodes usually focus only on their own responsibilities and rarely participate directly in other parts of the decision process.

As system scale increases, this differentiated structure becomes more complex. In large technical systems, data processing, algorithmic design, system integration, and business application are often handled by different teams. These teams coordinate through technical interfaces and information systems without directly participating in one another’s workflows.

In such environments, overall system behavior is jointly produced by multiple nodes, while each node assumes only local tasks. When consequences arise, different nodes may accept responsibility only for their own limited functions, while no single subject can fully explain the overall result.

This structural feature gradually shifts responsibility from concentrated states toward distributed states. Traditional responsibility structures usually assume that decision-makers can exercise overall control over system behavior, allowing responsibility to remain concentrated in one actor. In multi-node systems, however, overall behavior is produced by multiple interacting nodes. Different nodes assume local tasks, while the overall result is jointly formed.

As these structures expand, the degree of concentration of responsibility may gradually decline. Each node still bears some responsibility, but the concentrated position of responsibility within the system becomes harder to identify. When consequences occur, different nodes may explain only those actions that fall within their own domains, while overall system responsibility becomes difficult to attribute.

Responsibility fragmentation can therefore be understood as an intermediate state of responsibility structures. In this state, responsibility has not entirely lost its position, but it has already shifted from traditional concentration toward distribution. Different nodes bear local tasks, while overall behavior is jointly formed through them.

In AI-mediated systems, responsibility fragmentation is closely tied to technical structure. As AI increasingly participates in decision processes, the number of nodes within systems increases. Data systems, algorithmic models, software platforms, and organizational decision layers form complex interactions, causing responsibility structures to take on distributed characteristics.

Responsibility fragmentation does not mean the disappearance of responsibility. Rather, it indicates that the concentrated position of responsibility is gradually weakened. As systems continue to expand and automation increases, the state of fragmented responsibility may further evolve into responsibility drift.

Responsibility fragmentation can therefore be understood as an important stage in the formation of responsibility drift. By analyzing how responsibility becomes fragmented across multi-node systems, it becomes possible to understand more clearly how responsibility drift emerges in complex systems.

X. Mechanisms of Replication Expansion

Artificial intelligence systems possess strong replication capacity. Compared with traditional technical systems, algorithmic models, software systems, and data structures can be quickly extended to multiple environments through cloud platforms, API interfaces, and automated deployment mechanisms. In modern technical architectures, a single model can often run simultaneously across multiple system nodes and be repeatedly invoked in different business contexts.

This replication capacity allows technical systems to cover large application environments in relatively short periods of time. Models can be deployed across multiple server environments, software interfaces can be called by different systems, and automated workflows can operate simultaneously across multiple business nodes. As technical systems expand, outputs can continue to diffuse across organizations, regions, and even industries.

However, the replication of system behavior is not accompanied by the replication of responsibility structures. When algorithmic models or system components are repeatedly used in different environments, system behavior may diffuse across multiple nodes while responsibility structures do not necessarily extend in the same way. Technical systems can replicate quickly through software architectures, whereas responsibility structures still depend on organizational arrangements or decision subjects for confirmation.

This structural difference creates a form of **replication asymmetry**. Technical systems are capable of rapidly expanding system behavior, while responsibility structures are unable to scale at the same pace. As outputs continue to propagate across nodes, the concentrated position of responsibility may gradually weaken.

This phenomenon is especially visible in AI-mediated systems. Algorithmic models can be invoked simultaneously by multiple applications, and automated decision processes can run continuously in different organizational environments. As system behavior is replicated through technical structures, the range of system impact rapidly expands.

Responsibility structures, however, cannot usually replicate in the same way. Different organizations, teams, and technical nodes often bear only local functions, while overall system behavior is jointly produced. Under such conditions, the concentrated position of responsibility becomes increasingly blurred.

As replication capacity continues to grow, system behavior may keep diffusing across multiple environments. When system outputs are repeatedly reused, structural distance may emerge between the original decision structure and the actual

execution environment. Under these conditions, the concentrated position of responsibility may weaken further.

Replication expansion therefore constitutes an important driving force behind responsibility drift. When outputs can rapidly propagate across different nodes while responsibility structures cannot extend correspondingly, the degree of responsibility concentration gradually declines. As replication expansion continues, responsibility structures may gradually shift from concentrated to distributed states.

In complex technical systems, replication expansion interacts with responsibility fragmentation to drive changes in responsibility structure. When multi-node structures and replication capacity coexist, the concentrated position of responsibility may gradually shift, thereby producing responsibility drift.

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                AI Model Replication
      (Models deployed across multiple systems)

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             Execution Diffusion
   (Automated systems operating across contexts)

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                          ▼

              Impact Expansion
     (System outcomes spread across nodes)

                          |
                          ▼

             Responsibility Dilution
    (Responsibility structures gradually weaken)

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                          ▼

               Responsibility Drift
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    (Responsibility can no longer remain concentrated
                    in a single node)

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Figure 2. Responsibility Drift in Replicated AI Systems

In highly replicable AI systems, algorithmic models can operate across multiple environments through cloud services, API interfaces, or automated deployment. As execution capacities diffuse across multiple nodes, the range of system impact expands accordingly. Responsibility structures, however, do not replicate in the same way, which may lead responsibility to become diluted and drift within system structures.

Source: Author’s conceptual framework.

XI. Mechanisms of Responsibility Discontinuity

In highly automated systems, decision processes often take the form of continuous technical processing. A typical sequence includes data input, model computation, system recommendation, and automated execution. Under such conditions, information moves from the data layer into the algorithmic layer, is transformed into outputs through model computation, and then directly enters execution through automated processes.

Compared with traditional decision structures, this technical flow displays strong continuity. System behavior can be transmitted automatically across different technical layers without requiring human confirmation at each stage. System operation therefore exhibits a high degree of automation.

However, when outputs can directly enter execution, human actors may fail to assume responsibility-bearing roles at critical points. In traditional decision structures, decision-makers are typically required to confirm execution behavior, thereby forming a responsibility-bearing position. When consequences arise, responsibility can be identified through that position.

In automated environments, this responsibility-bearing position may gradually weaken. Outputs generated by algorithmic models can be automatically transmitted through software systems and continue to propagate across system nodes. When execution is largely completed by technical systems, human actors may participate only in design or oversight rather than in immediate responsibility-bearing.

Under such conditions, discontinuities may emerge in the responsibility chain. System behavior continues to operate, but responsibility-bearing positions may fail to form clearly within the decision chain. Outcomes are jointly produced by multiple nodes, while the concentrated position of responsibility becomes difficult to identify.

Responsibility discontinuity does not mean that system behavior lacks causal explanation. Rather, it means that continuity in the responsibility chain is interrupted. Different nodes still perform their respective functions, but continuity in the structural position of responsibility is no longer maintained. When outcomes arise, responsibility may be difficult to trace through any single node.

As automation intensifies, this condition may become even more pronounced. When outputs can enter execution directly through technical processes and human actors fail to assume responsibility at critical points, continuity in the structural location of responsibility gradually weakens.

In complex systems, responsibility discontinuity often works together with responsibility fragmentation and replication expansion. Multi-node structures gradually distribute responsibility, replication expansion propagates system behavior across multiple environments, and automation further weakens responsibility-bearing positions. When these factors occur simultaneously, continuity in responsibility structures is increasingly undermined.

Responsibility discontinuity therefore constitutes another important condition for responsibility drift. When responsibility-bearing relations within the decision chain cannot maintain continuity, the concentrated position of responsibility may gradually shift, giving rise to responsibility drift.

XII. Responsibility Drift and System Risk

Responsibility drift is not only a structural phenomenon, but may also have important implications for system operation. As the concentrated position of responsibility within the system gradually weakens, responsibility-bearing relations within decision chains become more difficult to identify. Under such conditions,

system behavior may continue to operate, while the stability of responsibility structures declines.

In complex organizational environments, responsibility structures are closely tied to governance structures. When responsibility retains a clear position within the system, organizations can typically coordinate and adjust through existing procedures. When responsibility drifts, however, governance may face new difficulties. Different nodes assume local duties while overall system behavior is jointly produced, making responsibility-bearing relations more complex.

Responsibility drift may also affect the identification of system risk. In traditional decision structures, risk is often assessed and managed through decision-bearing subjects. In multi-node systems, by contrast, behavior is jointly produced by multiple nodes, and each node may focus only on the risks within its own limited domain. When overall system risk is jointly formed through multiple local actions, risk identification becomes more difficult.

As AI systems increasingly participate in complex decision environments, this structural transformation may become more visible. Technical systems can rapidly propagate information and execution behavior across nodes, while organizational responsibility-bearing positions may fail to adjust correspondingly. Under such conditions, system behavior continues to expand while the degree of concentration of responsibility declines.

As responsibility structures move gradually from concentrated to distributed forms, decision stability may also be affected. Decision stability usually depends on the existence of clear responsibility-bearing positions that allow system behavior to be confirmed at key points. When such positions weaken, system operation may continue, but decision structures become increasingly complex.

Responsibility drift should therefore be understood not only as a structural transformation, but also as one factor in the evolution of system risk. As the concentrated position of responsibility weakens, decision chains may take on increasingly dispersed forms. Under such conditions, the relationship between system stability and responsibility structures warrants further study.

The concept of responsibility drift proposed here is intended to provide an analytical framework for understanding this structural transformation. By treating responsibility as a structural variable, it becomes possible to analyze from a complex systems perspective how responsibility structures shift as technical environments change.

XIII. Responsibility Structures and System Stability

In complex systems research, system stability is closely tied to system structure. Relations among nodes, modes of information flow, and feedback mechanisms all affect the long-term stability of system behavior. Within this theoretical framework, responsibility structures can be understood not only as governance issues, but also as important structural variables within systems.

In traditional organizational structures, responsibility typically corresponds to authority. Decision-makers possess authority and bear corresponding responsibility. When problems arise, responsibility can be identified and traced through organizational structure. This provides a stable responsibility-bearing mechanism, maintaining a clear relation between decision behavior and outcomes.

In complex systems, however, behavior is jointly produced through interactions among multiple nodes. As scale and structural complexity increase, node numbers grow and more intricate interaction networks form. Under such conditions, system stability often depends on the coordination of multiple structural variables.

Responsibility structures can be understood as one important component of such coordination. When responsibility remains relatively concentrated, responsibility-bearing relations within decision chains stay clear. Even if behavior is jointly produced by multiple nodes, critical decision points can still form stable positions within the system. Such structures help maintain continuity between system behavior and responsibility-bearing.

When responsibility structures gradually move from concentrated to distributed forms, however, system stability may be affected. In multi-node systems, different nodes assume only local tasks, while overall behavior is jointly formed. As the concentrated position of responsibility weakens, the responsibility-bearing relations within decision chains become more complex.

This transformation may be particularly visible in AI-mediated systems. Technical systems can rapidly process information and execute automated procedures, while organizational responsibility-bearing positions may fail to adjust correspondingly. As system behavior continues to propagate across nodes, responsibility structures may gradually shift from concentrated toward distributed forms.

When the degree of concentration of responsibility declines, system behavior may still continue, but the responsibility-bearing relations within decision chains become harder to identify. Under these conditions, stability may depend increasingly

on interactions between technical and organizational structures, while the function of responsibility within the system also changes.

From a complex systems perspective, responsibility structures are therefore not merely part of governance structures; they are also important variables in system stability. When the concentrated position of responsibility remains stable, system behavior and responsibility-bearing relations maintain a relatively high degree of coherence. When responsibility structures gradually drift, decision structures become correspondingly more complex.

By introducing the concept of responsibility drift, this paper brings responsibility structures into the analytical framework of complex systems stability. Treating responsibility as a structural variable makes it possible to analyze how responsibility structures shift in complex decision systems as technical environments change.

XIV. The Structural Role of Human Judgment

The present study builds upon the author's earlier theory of the **Human Judgment Interface**. In complex systems, human judgment is not merely one component of decision behavior; it constitutes an important interface within the system structure. That theory argues that in highly complex and only partially predictable environments, human judgment forms an irreducible interface position within the system.

In complex decision systems, technical systems can carry out tasks such as information processing, pattern recognition, and automated computation. However, when systems face highly uncertain environments, technical structures often cannot fully replace the role of human judgment. Human judgment introduces interpretation, weighing, and contextual understanding into system structures, allowing systems to retain structural stability under complexity.

Human judgment can therefore be understood as a structural interface in complex systems. This interface allows systems to preserve structural boundaries even as automated execution capacities increase. When systems enter highly automated environments, human judgment provides a responsibility-bearing and decision-bearing position within the structure.

In the author's earlier work, human judgment was treated as an **Irreducible Interface** in complex systems. This interface does not depend entirely on technical structures, but instead is constituted by human decision-making capacities within

the system. When behavior requires interpretation and balancing under uncertainty, human judgment forms a critical node.

This has important implications for the responsibility structures discussed here. When system behavior is jointly produced by multiple technical nodes, responsibility positions may gradually change. If systems lack clearly identifiable responsibility-bearing positions, responsibility structures may begin to drift.

When the human judgment interface is present, systems can form responsibility-bearing positions at key points in the decision chain. Such positions do not merely perform judgment functions; they also serve as important nodes in responsibility structures. The **Human Responsibility Node** proposed by the author is built precisely on this theoretical basis.

Human judgment therefore constitutes not only decision capacity, but also the structural position through which responsibility is borne. When human judgment occupies a clear place in the decision chain, responsibility can remain relatively concentrated. Conversely, when the position of human judgment is weakened, responsibility structures may move from concentrated toward distributed forms.

In AI-mediated environments, as automated decision capacities continue to expand, the position of human judgment may change. When technical nodes perform the bulk of system behavior, human judgment may be shifted backward in the decision chain. Under such conditions, responsibility-bearing positions become unstable, thereby creating structural conditions for responsibility drift.

The concept of responsibility drift thus stands in continuity with the Human Judgment Interface theory. Human judgment provides the structural interface for responsibility-bearing, while responsibility nodes describe the concentrated position of responsibility within the system. When this structural relation changes, the position of responsibility may gradually drift.

Relevant prior studies include:

Tu, X. (2026). *Judgment Before Momentum (1.0.2)*. Zenodo.

https://doi.org/10.5281/zenodo.18571480

Tu, X. (2026). *Human Judgment as an Irreducible Interface in High-Complexity Systems (1.0)*. Zenodo.

https://doi.org/10.5281/zenodo.18573944

XV. The Stabilizing Function of the Responsibility Node

In earlier work, the author proposed the concept of the **Human Responsibility Node** to describe a structural position in complex decision systems capable of bearing responsibility. Unlike traditional concepts of responsibility, the responsibility node is not understood primarily as an ethical or institutional arrangement in which one subject bears responsibility. Rather, it is treated as a positional variable within system structure.

In complex decision environments jointly constituted by human actors and technical systems, system behavior is often produced by interactions among multiple nodes. As systems expand and develop multi-node structures, the concentrated position of responsibility may gradually change. Under such conditions, if systems lack clear responsibility-bearing nodes, responsibility structures may gradually shift from concentrated to distributed modes.

The concept of the responsibility node is introduced precisely against this structural background. It describes a position in the decision chain capable of bearing responsibility. When system outputs enter execution, this node forms a responsibility-bearing relation within the decision chain, thereby maintaining the connection between system behavior and responsibility structure.

In complex system environments, the existence of responsibility nodes helps maintain decision-structural stability. Even when system behavior is jointly produced by multiple nodes, responsibility structures can remain relatively concentrated as long as clearly identifiable responsibility-bearing positions exist. The responsibility node can therefore be understood as an important interface connecting system behavior and responsibility structure.

As system structures expand, the function of the responsibility node becomes increasingly important. Multi-node decision structures may cause responsibility to fragment, while technical replication may further expand the range of system effects. Under such conditions, responsibility nodes form stable points within the decision chain, preventing responsibility structures from diffusing entirely.

A responsibility node does not imply that all decisions must be concentrated in a single subject. Rather, it emphasizes that complex system structures require key points capable of bearing responsibility. As system behavior continues to propagate through technical structures, such nodes form structural points of stability within the decision chain.

In AI-mediated systems, responsibility nodes typically appear at critical decision stages, such as system deployment decisions, model-application decisions, or execution confirmations. These positions allow responsibility-bearing relations to be formed before system behavior enters actual execution, thereby maintaining continuity in responsibility structures.

The responsibility node can therefore be understood as a mechanism of structural stability in complex systems. When responsibility nodes remain stable, responsibility structures can remain relatively concentrated. Conversely, when systems lack clear responsibility nodes, the concentrated position of responsibility may gradually shift, giving rise to responsibility drift.

Within the theoretical framework developed here, the human judgment interface, responsibility node, and responsibility drift form a continuous structural relation. Human judgment provides the decision interface, the responsibility node forms the responsibility-bearing position, and responsibility drift describes the displacement of responsibility when these structural relations are transformed.

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                Human Judgment

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        Human Responsibility Node

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        Responsibility Stability

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           Structural Stability

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   Human Judgment in Complex Systems Research
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Figure 3. Human Judgment and Responsibility Structure

In complex decision systems jointly constituted by humans and artificial intelligence, human judgment functions as a structural interface within the system. When clear responsibility nodes are present, responsibility structures can remain concentrated within the decision chain. The existence of responsibility nodes helps maintain responsibility-bearing relations, thereby improving overall structural stability.

Source: Author's conceptual framework.

XVI. Responsibility Issues in AI Governance

As AI systems become increasingly embedded in social operation, technical systems participate in ever more important decision processes. From enterprise operations and financial decisions to infrastructure management and public service systems, AI technologies are becoming key components of complex decision environments. Against this backdrop, responsibility structures are gradually becoming an important issue in AI governance research.

In traditional governance structures, responsibility generally corresponds to organizational authority. Decision-bearing subjects possess decision power and assume corresponding responsibility. When system behavior generates consequences, responsibility can be identified and traced through organizational structure. In industrial and early information-technology settings, such structures were relatively stable.

However, as AI increasingly participates in decision processes, system structures change. Decision chains are no longer composed solely of human actors, but rather of technical and organizational nodes. Data systems, algorithmic models, software platforms, and organizational decision layers form complex interactive relations, producing distinct multi-node decision structures.

In such AI-mediated systems, system behavior is typically jointly produced by multiple nodes. Technical systems handle information processing and automated execution, while organizational structures apply outputs in concrete decision environments. As system complexity increases, the concentrated position of responsibility may gradually shift.

AI governance research often focuses on algorithmic transparency, system explainability, and technical regulation. These approaches are important for understanding how AI systems operate. Yet within complex systems environments, responsibility structure itself may also become an important structural variable.

When system behavior is jointly produced by multiple nodes, responsibility no longer remains stably concentrated in one subject. Different nodes assume local responsibilities, while overall system behavior emerges through their combination. Under such conditions, the position of responsibility may gradually shift from concentrated toward distributed forms.

As the replication capacity and automation capacity of technical systems continue to increase, system behavior can expand rapidly across nodes. Responsibility structures, however, may not expand correspondingly. When system behavior continues to propagate while responsibility-bearing positions fail to form in parallel, the degree of concentration of responsibility may gradually decline.

Thus, in AI-mediated environments, responsibility problems concern not only institutional arrangements, but also system structure itself. The position, concentration, and bearing of responsibility may all shift as technical structures change.

The concept of responsibility drift proposed here is intended to provide an analytical framework for understanding such structural transformation. By treating responsibility as a structural variable in complex systems, it becomes possible to analyze the pathways through which responsibility structures change in AI environments.

This perspective does not directly prescribe governance measures. Rather, it seeks to describe the forms that responsibility structures may take in AI-mediated systems. As AI technologies continue to develop, understanding these structural changes becomes important for analyzing the long-term stability of complex decision systems.

XVII. Relations Within the Theoretical System

This study forms part of the author's broader theoretical system on **Human Judgment in Complex Systems**. That research program concerns the structural relations among human judgment, responsibility structures, and system stability in complex technological environments. As AI systems become more deeply embedded

in social operation, traditional decision structures change, and the relation between human actors and technical systems becomes an increasingly important issue in complex systems research.

Within this theoretical system, human judgment is treated as an important structural variable. Technical systems can perform tasks such as information processing, pattern recognition, and automated computation, but under conditions of high uncertainty, human judgment continues to play a critical structural role. Introducing human judgment into structural analysis makes it possible to understand how system stability may be maintained as automation continues to expand.

The author's earlier work introduced **Judgment Before Momentum**, emphasizing that judgment must form structural boundaries before execution momentum expands. That study argued that as execution capacity increases, judgment becomes especially important in preserving interpretability and structural stability.

On this basis, the author further developed the theory of **Human Judgment as an Irreducible Interface**. This work argued that human judgment can be understood as an irreducible interface in complex technical systems. When systems face highly uncertain environments, human judgment provides the interface through which interpretation and balancing occur, enabling structural boundaries to be retained despite expanding automation.

The subsequent concept of the **Human Responsibility Node** described a structural position within complex decision systems capable of bearing responsibility. The responsibility node emphasized that even within multi-node decision structures, key responsibility-bearing positions must remain present in order to preserve the connection between system behavior and responsibility structure. This concept transformed responsibility from a primarily normative notion into a positional variable within system structure.

Building on these studies, the present paper introduces **Responsibility Drift** as a concept describing the diffusion and displacement of responsibility in AI-mediated systems. When system behavior is jointly produced by multiple nodes and responsibility-bearing positions fail to remain stable, the concentration of responsibility may gradually decline, thereby producing responsibility drift.

The present study can therefore be understood as an important part of this broader theoretical system. These four studies together form a continuous framework:

Human Judgment as an Irreducible Interface describes the structural interface function of human judgment in complex systems.

Human Responsibility Node describes the responsibility-bearing position within decision chains.

Responsibility Drift describes the displacement of responsibility when these structural relations are transformed.

Through this theoretical system, it becomes possible to analyze from a complex systems perspective the relations between human judgment and responsibility structures in AI environments, and to better understand the structural conditions of system stability in automated decision contexts.

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Human Judgment Interface

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Human Responsibility Node

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Responsibility Stability

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Responsibility Drift

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Structural Change

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Human Judgment in Complex Systems Research
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Figure 4. Human Judgment–Responsibility Framework

In complex systems jointly constituted by human actors and artificial intelligence, human judgment functions as a structural interface. When stable responsibility nodes are present, responsibility can remain concentrated in the decision chain, thereby maintaining structural stability. When automation continues to increase and responsibility nodes gradually weaken, responsibility structures may begin to drift. Responsibility drift further affects governance structure and may contribute to broader structural change.

Source: Author's conceptual framework.

XVIII. Conclusion

This paper introduces the concept of **Responsibility Drift** to describe the structural phenomenon through which responsibility in AI-mediated systems gradually shifts from concentrated to distributed states. As AI systems become more widely embedded in organizational governance, business decision-making, and socio-technical infrastructures, decision processes are increasingly completed by multi-node structures. Under such conditions, system behavior is no longer directly generated by single subjects, but jointly formed through interactions among technical and organizational nodes.

In traditional decision structures, responsibility can usually be identified through organizational hierarchy. Decision subjects and execution behavior maintain relatively clear correspondences, allowing responsibility to remain concentrated. In AI-mediated systems, however, data systems, algorithmic models, software architectures, and organizational decision layers form complex interactions, producing increasingly distributed decision structures.

Through the analysis of multi-node structures, replication expansion, and automated execution, this paper has shown how responsibility structures may gradually change in complex technical environments. When system behavior is jointly produced by multiple nodes and responsibility-bearing positions fail to remain stable, the degree of concentration of responsibility may decline, thereby giving rise to responsibility drift.

The paper further argues that responsibility drift does not mean the disappearance of responsibility. Rather, it means that the position of responsibility within the system changes. In complex technical environments, responsibility can be understood as a structural variable. As systems continue to expand and automation intensifies, responsibility structures may move gradually from traditional concentrated forms toward distributed forms.

Within this framework, the human judgment interface, the human responsibility node, and responsibility drift form a continuous structural relation. Human judgment functions as an irreducible interface, responsibility nodes provide responsibility-bearing positions within decision chains, and responsibility drift describes the displacement of responsibility when these structural relations are transformed. This perspective helps explain how responsibility structures change as technical systems expand.

This study attempts to move the problem of responsibility from traditional normative discussion into the structural analytical framework of complex systems. By treating responsibility as a structural variable, it becomes possible to analyze from a complex systems perspective the pathways through which responsibility structures change in AI-mediated environments. This perspective is valuable for understanding the relation between responsibility structures and system stability in automated decision settings.

As AI systems continue to expand in social operation, responsibility structures in complex decision systems are likely to become an increasingly important research topic. The concept of responsibility drift proposed here provides an initial structural framework for analyzing this phenomenon and offers a theoretical basis for future research on responsibility structures in complex systems.

Conclusion

This study contributes to the understanding of responsibility structures in AI-mediated systems by introducing the concept of Responsibility Drift as a structural variable within complex decision architectures.

These works provide the theoretical foundation for the Human Responsibility Node and Responsibility Drift framework discussed in this paper.

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